

强化学习 (RL) 最新综述调研报告 (截至 2026-03-03)

生成时间: 2026-03-03

调研范围: 优先 arXiv 的 RL 及 RL+子方向综述 (survey/review/technical survey)

目标: 筛选当前最新且有代表性的综述, 并给出可执行阅读路线

一、关键结论

- 1) 最新已进入 2026 年批次: 截至 2026-03-03, 已出现 2026-01 和 2026-02 的 RL 综述 (real-world RL、time-delay control RL、safe continual RL)。
- 2) 主流热点仍在 LLM/Agentic RL: 2025 下半年出现大量 RL-for-LLM/LRM/Agent 综述, 并在 2026
- 3) 安全与持续学习方向升温: Safe RL + Continual RL 交叉综述开始出现, 强调 nonstationary 环境
- 4) 通用入门与前沿并进建议: 通用综述 1 篇 + 2026 新综述 2 篇 + 领域专题 2 篇。

二、最新综述清单 (按时间优先)

A. 2026 (当前最“新”)

- 1) Statistical Reinforcement Learning in the Real World: A Survey of Methods and Open Challenges
arXiv:2601.15353 (2026-01)
- 2) Safe Continual Reinforcement Learning Methods for Nonstationary Environments
arXiv:2601.05152 (2026-01-08)
- 3) Reinforcement Learning for Control Systems with Time Delays: A Survey
arXiv:2602.00399 (2026-02)

B. 2025 (近期高影响专题)

- 4) A Survey of Reinforcement Learning for Large Reasoning Models
arXiv:2509.08827
- 5) The Landscape of Agentic Reinforcement Learning for LLMs: A Survey
arXiv:2509.02547
- 6) Reinforcement Learning Meets Large Language Models
arXiv:2509.16679
- 7) Reinforcement Learning for Large Model: A Survey
arXiv:2508.08189
- 8) A Survey of Continual Reinforcement Learning
arXiv:2506.21872
- 9) A Survey of Safe Reinforcement Learning and Constrained MDPs
arXiv:2505.17342
- 10) A Survey of In-Context Reinforcement Learning
arXiv:2502.07978

C. 2024 (基础参照)

- 11) A Comprehensive Survey of Reinforcement Learning
arXiv:2411.18892
- 12) Reinforcement Learning Enhanced LLMs: A Survey
arXiv:2412.10400
- 13) A Survey of Constraint Formulations in Safe Reinforcement Learning
arXiv:2402.02025

三、趋势总结

- 1) 从“算法大全”转向“系统问题”: 更强调训练基础设施、数据构造、评测协议、部署成本。
- 2) RL 与 LLM 深度耦合: RLVR、Agentic RL、长链路推理成为后训练核心机制。
- 3) 安全与真实部署并重: safe continual、real-world gap、time-delay control 成为新焦点。

四、建议阅读路径

- 1) 先建框架: 2411.18892
- 2) 补前沿: 2509.08827 + 2509.02547 + 2509.16679
- 3) 补落地: 2505.17342 + 2601.15353 + 2602.00399